# STANDATA bulletin 23-BCB-006

Building

## SEER2 and HSPF2 application

Date Issued: March 2025



### Purpose

This bulletin provides information on the latest energy efficiency metrics for air-cooled unitary air conditioners and heat pumps that are electrically operated.

### Discussion

Energy efficiency requirements for mechanical equipment are readily changing as Canada aims to reduce greenhouse gas emissions.

The National Building Code – 2023 Alberta Edition (NBC(AE)) and the National Energy Code of Canada for Buildings 2020 (NECB) outline the minimum performance requirements for heating, ventilation, and air conditioning (HVAC) equipment. These performance requirements are based on the energy efficiency of the HVAC equipment. For electrically-operated, air-cooled unitary air conditioners and heat pumps, the energy efficiency ratio (EER) is used as a steady-state metric, while the seasonal energy efficiency ratio (SEER) and heating seasonal performance factor (HSPF) are used as seasonal performance metrics in the NBC(AE) and NECB. See Appendix A for more information regarding these performance metrics.

A new test procedure (Appendix M1 to Subpart B of Part 430 of the United States Code of Federal Regulations, entitled Uniform Test Method for Measuring the Energy Consumption of Central Air Conditioners and Heat Pumps ("M1")) has been adopted by Canada into the Energy Efficiency Regulations (Canada). This has resulted in new seasonal performance metrics that are to be used for single-phase residential air conditioners and heat pumps with a cooling or heating capacity of less than 19 kW (65,000 Btu/h). EER2, SEER2, and HSPF2 are the new metrics that will be discussed in this bulletin.

The Energy Efficiency Regulations now require single-phase residential air conditioners and heat pumps with a cooling or heating capacity of less than 19 kW (65,000 Btu/h), manufactured after January 1, 2023, and imported into Canada or shipped between provinces for the purpose of sale, to be tested to the new M1 test procedure with HSPF2 calculated and reported for Climate Region V (i.e. region five) (see Appendix B). As the industry transitions to the new M1 test procedure, testing of equipment at the very low temperature point (-15$^{0}$C) will be optional until January 1$^{st}$, 2025. After that, it will become mandatory to test and report equipment performance at the very low temperature point. Given the demand for this type of equipment, it is expected that existing stock rated under the old test procedure (CSA C656-14 for seasonal energy efficiency ratio and heating seasonal performance factor) will be depleted quickly, meaning that minimum performance requirements stated in HSPF, EER and SEER will soon become obsolete.

It is important to note that direct conversions from EER, SEER, and HSPF to EER2, SEER2, and HSPF2 are not possible due to the fundamental differences in the testing procedures.



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Table 9.36.3.10. of the NBC(AE) and Table 5.2.12.1.-A of the NECB provide the minimum SEER2 and HSPF2 V (heating seasonal performance factor 2 for region V) requirements for electrically-operated, air-cooled unitary air conditioners and heat pumps.

Natural Resources Canada (NRCan) provides a database of approved equipment that meets the minimum energy efficiency metrics. This database can be accessed at the following link:
https://oee.nrcan.gc.ca/pml-lmp/index.cfm?action=app.welcome-bienvenue

HVAC equipment products, including heat pumps, are regulated products in Canada, which means they cannot be imported to Canada or shipped between provinces for the purpose of sale unless they meet a minimum efficiency standard. As such, it is recommended to ensure that heat pumps being installed in Canada are models that have been approved by NRCan and exist in their database.

Note that climate regions and climate zones are similar in concept but have different applications and therefore should not be used interchangeably. Climate zones, as referenced in the NBC(AE), are geographical areas delineated based on heating degree days (HDD). Climate regions, as referenced in CSA C656-14 and “M1,” are geographical areas that have been delineated based on design temperature and heating load hours below 18°C and are used for testing HVAC equipment.

## Code References

**Article 9.36.3.10.**

**9.36.3.10. Equipment Efficiency**
1) HVAC equipment and components shall comply with the performance requirements stated in Table 9.36.3.10. (See Note A-9.36.3.10.(1).)

**A-9.36.3.10.(1) Unit and Packaged Equipment.** The minimum performance values stated in Table 9.36.3.10. were developed based on values and technologies found in the Model National Energy Code of Canada for Houses 1997, the NECB, federal, provincial and territorial energy efficiency regulations as well as in applicable standards on equipment typically installed in housing and small buildings.
In some cases—after a review of current industry practices (industry sales figures)—the performance requirements were increased from regulated minimums where it could be shown that the cost and availability of the equipment are acceptable. Some of the performance requirements are based on anticipated efficiency improvements in the energy efficiency regulations and revisions to standards.

**Table 9.36.3.10.**

**Table 9.36.3.10.**
**HVAC Equipment Performance Requirements**
Forming Part of Sentences 9.36.3.9.(2) and 9.36.3.10.(1)

| Type of Equipment | Heating or Cooling Capacity, kW | Performance Testing Standard | Minimum Performance(1) |
|---|---|---|---|
| **Air-Cooled Unitary Air Conditioners and Heat Pumps – Electrically Operated** | | | |
| Split system | < 19 | CSA C656 | SEER = 14.5<br>EER = 11.5<br>HSPF V = 7.1 |
| | | DOE 10 CFR, Part 430, Subpart B, Appendix M1 | SEER2 = 14.3<br>HSPF2 V = 6.0 |
| Single-package system | < 19 | CSA C656 | SEER = 14<br>EER = 11<br>HSPF V = 7.0 |
| | | DOE 10 CFR, Part 430, Subpart B, Appendix M1 | SEER2 = 13.4<br>HSPF2 V = 5.4 |



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**Notes to Table 5.2.12.1.-A:**

(1) The symbols and abbreviations that appear in this column have the following meanings:
EER = energy-efficiency ratio, in (Btu/h)/WEER = energy-efficiency ratio, in (Btu/h)/W
HSPF V = heating seasonal performance factor for region V (see map in CSA C656), in (Btu/h)/W
HSPF2 V = heating seasonal performance factor 2 for region V (see map in DOE 10 CFR, Part 430, Subpart B), in (Btu/h)/W
SEER = seasonal energy-efficiency ratio, in (Btu/h)/W
SEER2 = seasonal energy-efficiency ratio 2, in (Btu/h)/W

**NECB 2020, Division B, Article 5.2.12.1.**

**5.2.12.1. Unitary and Packaged HVAC Equipment**
1) Unitary and packaged HVAC equipment and components with the capacities listed in Tables 5.2.12.1.-A to 5.2.12.1.-P shall comply with the performance requirements stated therein. (See Notes A-5.2.12.1.(1) and A-5.2.12.1.(1) and 6.2.2.1.(1).) (See also Article 6.2.2.4.)

**A-5.2.12.1.(1) Unitary and Packaged HVAC Equipment.** Units of equipment subject to federal, provincial or territorial appliance or equipment energy efficiency acts carry a label certifying that their performance meets the requirements of the standard and acts shown thereon; there is therefore no need for figures to be checked.
In cases where third-party certification is not feasible (due to the absence or limitations of certification programs or testing facilities), the equipment's performance is usually demonstrated with data that is obtained in accordance with the referenced testing standard and supplied by the manufacturer.
It should be noted that, where a building is served by multiple heating or cooling units that are activated in sequence in response to increasing heating or cooling needs, it is likely economically justified to specify higher efficiency than is mandated in this Code for the lead units, which operate for the longest periods of time.

**A-5.2.12.1.(1) and 6.2.2.1.(1) Performance Requirements and Levels.**
**Performance Requirements**
HVAC and service water heating equipment standards are reviewed and updated on a regular basis, whereas the "Energy Efficiency Regulations" are revised or updated to include new types of equipment at irregular intervals. The regulations follow a legislative protocol prior to becoming a federal requirement. This means that the publication of revisions to these documents does not always coincide with the publication of a new edition of the Code. As such, the performance requirement of any equipment or component in Tables 5.2.12.1.-A to 5.2.12.1.-P and 6.2.2.1. can change without notice between Code cycles.

**Performance Levels**
The federal "Energy Efficiency Act," which was introduced in 1992, provides for the development and enforcement of regulations concerning minimum energy performance levels for energy-using products and products that affect energy use, as well as the labeling of energy-using products and the collection of data on energy use.
The "Energy Efficiency Regulations," which came into effect in 1995, establish energy efficiency standards for a wide range of energy-using products imported into Canada or manufactured in Canada with the objective of eliminating the least energy-efficient products from the Canadian market. They set test procedures and require that each product carry a verification mark from a product certification body accredited by the Standards Council of Canada, which certifies that the product's energy performance is in compliance with the Regulations' energy efficiency standard for that type of product. The Regulations are amended on a regular basis in accordance with the federal government's regulatory process; a summary of the current Regulations is available at www.nrcan.gc.ca/energy-efficiency/energy-efficiencyregulations/%20guide-canadas-energy-efficiency-regulations/6861.



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**Table 5.2.12.1. -A**
**Performance Requirements for Air-Cooled Unitary Air Conditioners and Heat Pumps – Electrically Operated[1]**
Forming Part of Sentences 5.2.12.1.(1), 6.2.2.4.(2), 6.2.2.5.(1) and 8.4.4.18.(6)

| Type of Equipment | Cooling or Heating Capacity, kW | Performance Testing Standard | Minimum Performance[2] |
|---|---|---|---|
| Single-package, space-constrained | < 19 | CSA C656<br>or DOE 10 CFR Appendix M1 for EER2, SEER2 and HSPF2 | SEER = 13 / HSPF V = 6.4[3]<br>Or<br>SEER2 = 11.9 / HSPF2 V = 5.0 |
| Single-package, others | | | SEER = 15 / HSPF V = 7.4[3]<br>Or<br>SEER2 = 13.4 / HSPF2 V = 5.4 |
| Split system, space-constrained | | | SEER = 13 / HSPF V = 6.4[3]<br>Or<br>SEER2 = 11.9 / HSPF2 V = 5.0 |
| Split system, others | | | SEER = 15 / HSPF V = 7.4[3]<br>Or<br>SEER2 = 14.3 / HSPF2 V = 6.0 |
| Small-duct, high-velocity | | | SEER = 13 / HSPF V = 5.9[3]<br>Or<br>SEER2 = 12 / HSPF2 V = 4.9 |

**Notes to Table 5.2.12.1.-A:**

[1] Components or equipment regulated in the "Energy Efficiency Regulations" at the time of publication of the Code (see Article 1.1.1.3. of Division A).

[2] The symbols and abbreviations that appear in this column have the following meanings:

COPh = coefficient of performance in heating mode, in W/W
EER = energy-efficiency ratio, in (Btu/h)/W
HSPF V = heating seasonal performance factor for region V (see map in CSA C656), in (Btu/h)/W
HSPF2 V = heating seasonal performance factor 2 for region V (see map in DOE 10 CFR, Part 430, Subpart B), in (Btu/h)/W
IEER = integrated energy-efficiency ratio, in (Btu/h)/W
SEER = seasonal energy-efficiency ratio, in (Btu/h)/W
SEER2 = seasonal energy-efficiency ratio 2, in (Btu/h)/W

[3] SEER applies to air conditioners, and both SEER and HSPF V apply to heat pumps.

As the current energy code in force does not contain information on SEER2 and HSPF2 V, Table 5.2.12.1.-A and Notes to Table 5.2.12.1.-A in the code references above contains red text that indicates proposed changes to the NECB 2025 provided by NRCan.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition

Issued by the Provincial Building Administrator

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## APPENDIX A: Efficiency Terminology Provided by NRCan

**Steady-State Metrics:** These measures describe heat pump efficiency in a 'steady-state,' i.e., without real-life fluctuations in season and temperature. As such, their value can change significantly as source and sink temperatures, and other operational parameters, change. Steady state metrics include:

- **Coefficient of Performance (COP):** The COP is a ratio between the rate at which the heat pump transfers thermal energy (in kW), and the amount of electrical power required to do the pumping (in kW). For example, if a heat pump used 1kW of electrical energy to transfer 3 kW of heat, the COP would be 3.
- **Energy Efficiency Ratio (EER):** The EER is similar to the COP and describes the steady-state cooling efficiency of a heat pump. It is determined by dividing the cooling capacity of the heat pump in Btu/h by the electrical energy input in Watts (W) at a specific temperature. EER is strictly associated with describing the steady-state cooling efficiency, unlike COP which can be used to express the efficiency of a heat pump in heating as well as cooling.
- **Seasonal Performance Metrics:** These measures are designed to give a better estimate of performance over a heating or cooling season, by incorporating "real life" variations in temperatures across the season.

**Seasonal metrics include:**

- **Heating Seasonal Performance Factor (HSPF):** HSPF is a ratio of how much energy the heat pump delivers to the building over the full heating season (in Btu), to the total energy (in Watt-hours) it uses over the same period. Weather data characteristics of long-term climate conditions are used to represent the heating season in calculating the HSPF. However, this calculation is typically limited to a single region, and may not fully represent performance across Canada. Some manufacturers can provide an HSPF for another climate region upon request; however typically HSPFs are reported for Region 4, representing climates similar to the Midwestern US. Region 5 would cover most of the southern half of the provinces in Canada, from the B.C interior through New Brunswick[1].
- **Seasonal Energy Efficiency Ratio (SEER):** SEER measures the cooling efficiency of the heat pump over the entire cooling season. It is determined by dividing the total cooling provided over the cooling season (in Btu) by the total energy used by the heat pump during that time (in Watt-hours). The SEER is based on a climate with an average summer temperature of 28°C.

**Footnotes**

1. The Region 5 HSPF is most reflective of heat pump performance in the Ottawa region. Actual HSPFs may be lower in regions with increased heating degree days. While many colder Canadian regions are still classified under Region 5, the HSPF value provided may not fully reflect actual system performance.

The information provided in Appendix A can be found at the following link:
https://natural-resources.canada.ca/energy-efficiency/energy-star-canada/about/energy-star-announcements/publications/heating-and-cooling-heat-pump/6817

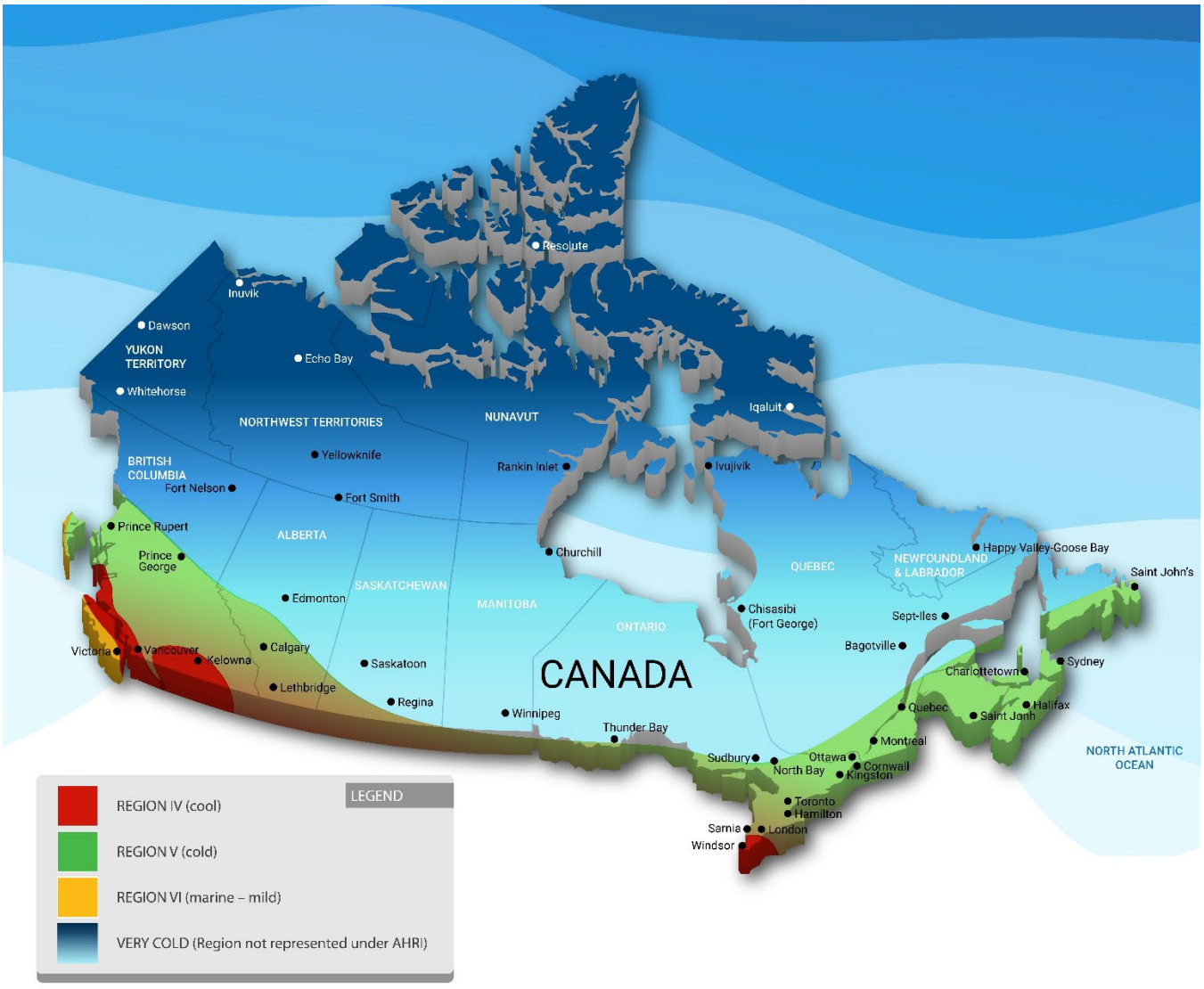


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## APPENDIX B: Climate Region Map

This map was created to provide a representation of the Air-Conditioning, Heating, and Refrigeration Institute (AHRI) regions for Canada. The AHRI regions were derived based on design temperature and heating load hours below 18°C.

The heat pump seasonal efficiencies will vary based on the climate zone and design temperature. Heating seasonal performance factors (HSPF) of the heat pumps for the AHRI regions may not be representative of actual performance. Further analysis is required to provide a more accurate estimate of potential energy savings.






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## APPENDIX C: Toolkit for Air Source Heat Pump Sizing and Selection

NRCan has developed a package of materials related to air source heat pump (ASHP) sizing and selection, intended for use by mechanical system designers and renovation contractors. The materials were designed to assist these individuals with sizing and selecting ASHPs for Canadian climates, in both new and existing (retrofit) residential applications.

**Components of the toolkit:**

1. **ASHP Sizing and Selection Guide** (referred to as “the Guide”) and **ASHP Key Specifications Summary Worksheet** (included in the Guide as Appendix B).
2. **ASHP Sizing and Selection Tool**: an electronic representation of the ASHP Sizing and Selection Guide. The tool interactively guides users in applying the step-by-step process described in the Guide, while providing additional charting and performance analysis features **(version V2.4 is now available. Existing users are encouraged to upgrade to V2.4)**.
3. **Addendum of worked examples**: provides case studies of the Guide being used to select either centrally ducted or ductless mini-split ASHPs for various installation scenarios.

The toolkit can be found at the following link:
https://natural-resources.canada.ca/maps-tools-and-publications/tools/modelling-tools/toolkit-for-air-source-heat-pump-sizing-and-selection/23558



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